Machine Learning Questions - Detailed Answers

Q20: Apply FIND-S Algorithm for Finding the Most Specific Hypothesis

FIND-S Algorithm Overview: The FIND-S algorithm finds the most specific hypothesis that is consistent with the positive training examples. It starts with the most specific hypothesis and generalizes it only when necessary to cover positive examples.

Algorithm Steps:

1. Initialize hypothesis h to the most specific hypothesis (all attributes set to null/∅)
2. For each positive training example:
    - If the example is positive, generalize h to cover the example
    - If attribute value in h is more specific than in example, replace it with the example's value
    - If attribute value in h is ∅, replace it with the example's value
3. Return the final hypothesis

Example Application: Given training data for concept "Enjoy Sport":

- Sky: Sunny, Cloudy, Rainy
- AirTemp: Warm, Cold
- Humidity: Normal, High
- Wind: Strong, Weak
- Water: Warm, Cool
- Forecast: Same, Change

Training Examples:

1. (Sunny, Warm, Normal, Strong, Warm, Same) → Yes
2. (Sunny, Warm, High, Strong, Warm, Same) → Yes
3. (Rainy, Cold, High, Strong, Warm, Change) → No
4. (Sunny, Warm, High, Strong, Cool, Change) → Yes

Step-by-step execution:

- Initial h = (∅, ∅, ∅, ∅, ∅, ∅)
- After example 1: h = (Sunny, Warm, Normal, Strong, Warm, Same)
- After example 2: h = (Sunny, Warm, ?, Strong, Warm, Same)
- Skip example 3 (negative)
- After example 4: h = (Sunny, Warm, ?, Strong, ?, ?)

Final Hypothesis: h = (Sunny, Warm, ?, Strong, ?, ?)

Q21: Compare and Contrast Supervised, Unsupervised, and Reinforcement Learning

Supervised Learning

Definition: Learning with labeled training data where the algorithm learns to map inputs to known outputs.

Characteristics:

- Uses labeled datasets
- Goal is to predict outcomes for new data
- Performance can be measured against known correct answers
- Examples: Classification and Regression

Applications:

- Email spam detection
- Medical diagnosis
- Stock price prediction
- Image recognition

Unsupervised Learning

Definition: Learning from data without labeled examples, finding hidden patterns or structures.

Characteristics:

- No labeled training data
- Goal is to discover hidden patterns
- More exploratory in nature
- Examples: Clustering and Association Rules

Applications:

- Customer segmentation
- Market basket analysis
- Anomaly detection
- Data compression

Reinforcement Learning

Definition: Learning through interaction with an environment, using rewards and penalties to improve decision-making.

Characteristics:

- Learning through trial and error
- Uses reward/penalty system
- Agent-environment interaction
- Goal is to maximize cumulative reward

Applications:

- Game playing (Chess, Go)
- Autonomous vehicles
- Robotics
- Trading algorithms

Key Differences:

Aspect	**Supervised**	**Unsupervised**	**Reinforcement**
Data Type	Labeled	Unlabeled	Interactive
Learning Goal	Prediction	Pattern Discovery	Optimal Actions
Feedback	Immediate	None	Delayed Rewards
Examples	Classification, Regression	Clustering, PCA	Q-learning, Policy Gradient

Q41: PCA Dimension Reduction - Dataset 1

Given Dataset: (2, 3), (3, 5), (5, 8), (8, 12), (10, 15)

Step 1: Calculate Mean

- Mean of $X_1 = (2+3+5+8+10)/5 = 5.6$
- Mean of $X_2 = (3+5+8+12+15)/5 = 8.6$

Step 2: Center the Data Subtract mean from each data point:

- (-3.6, -5.6), (-2.6, -3.6), (-0.6, -0.6), (2.4, 3.4), (4.4, 6.4)

Step 3: Calculate Covariance Matrix

$Cov(X_1,X_1) = \Sigma(x_{1i} - \mu_1)^2 / (n-1) = 102.8/4 = 25.7$

$Cov(X_2,X_2) = \Sigma(x_{2i} - \mu_2)^2 / (n-1) = 153.2/4 = 38.3$

$Cov(X_1,X_2) = \Sigma(x_{1i} - \mu_1)(x_{2i} - \mu_2) / (n-1) = 126.4/4 = 31.6$

Covariance Matrix:

C = [25.7 31.6]

[31.6 38.3]

Step 4: Calculate Eigenvalues Characteristic equation: $\det(C - \lambda I) = 0$ $\lambda^2 - 64\lambda + 23.81 = 0$ $\lambda_1 = 63.6$, $\lambda_2 = 0.4$

Step 5: Calculate Eigenvectors For $\lambda_1 = 63.6$: $v_1 = [0.624, 0.781]$ For $\lambda_2 = 0.4$: $v_2 = [-0.781, 0.624]$

Step 6: Project Data onto First Principal Component Principal Component 1 = $0.624 \times (X_1 - 5.6) + 0.781 \times (X_2 - 8.6)$

Transformed Data (1D):

- Point 1: -6.62
- Point 2: -4.44
- Point 3: -0.84
- Point 4: 4.16
- Point 5: 7.74

Q42: PCA Dimension Reduction - Dataset 2

Given Dataset: (3, 4), (2, 5), (5, 6), (7, 10), (10, 14)

Step 1: Calculate Mean

- Mean of $X_1 = (3+2+5+7+10)/5 = 5.4$
- Mean of $X_2 = (4+5+6+10+14)/5 = 7.8$

Step 2: Center the Data

- (-2.4, -3.8), (-3.4, -2.8), (-0.4, -1.8), (1.6, 2.2), (4.6, 6.2)

Step 3: Calculate Covariance Matrix

$Cov(X_1,X_1) = 34.8/4 = 8.7$

$Cov(X_2,X_2) = 68.8/4 = 17.2$

$Cov(X_1,X_2) = 48.4/4 = 12.1$

Covariance Matrix:

C = [8.7   12.1]

   [12.1 17.2]

Step 4: Calculate Eigenvalues $\lambda_1 = 24.5$, $\lambda_2 = 1.4$

Step 5: Calculate Eigenvectors For $\lambda_1 = 24.5$: $v_1 = [0.566, 0.824]$

Step 6: Project Data onto First Principal Component Transformed Data (1D):

- Point 1: -4.49
- Point 2: -4.24

- Point 3: -1.71
- Point 4: 2.72
- Point 5: 7.72

Q43: Chi-Square Test for Gender and Political Party Preference

Note: The specific table mentioned in the question is not provided in the PDF. I'll explain the general process for chi-square test.

Chi-Square Test Process:

Step 1: Set up Hypotheses

- H_0: Gender and political party preference are independent
- H_1: Gender and political party preference are dependent

Step 2: Calculate Expected Frequencies For each cell: Expected = (Row Total × Column Total) / Grand Total

Step 3: Calculate Chi-Square Statistic $\chi^2 = \Sigma[(\text{Observed} - \text{Expected})^2 / \text{Expected}]$

Step 4: Compare with Critical Value

- Degrees of freedom = (rows - 1) × (columns - 1) = 2
- Critical value at $\alpha = 0.05$ is 5.991

Step 5: Decision Rule

- If $\chi^2 > 5.991$, reject H_0 (variables are dependent)
- If $\chi^2 \leq 5.991$, fail to reject H_0 (variables are independent)

Interpretation: The chi-square test helps determine if there's a significant association between gender and political party preference by comparing observed frequencies with expected frequencies under the assumption of independence.

Q54: Linear Regression - Midterm and Final Exam Marks

Note: The specific table with midterm and final exam marks is referenced but not provided in the PDF. I'll demonstrate the general process.

Linear Regression Process:

Step 1: Set up the Problem

- Independent variable (X): Midterm marks
- Dependent variable (Y): Final exam marks

Step 2: Calculate Regression Coefficients For simple linear regression: $Y = a + bX$

Where:

- $b = \Sigma[(X - \bar{X})(Y - \bar{Y})] / \Sigma[(X - \bar{X})^2]$

- $a = \bar{Y} - b\bar{X}$

Step 3: General Formula Once calculated with actual data, the equation would be: Final Marks = a + b × Midterm Marks

Step 4: Prediction For midterm marks = 40: Final Marks = a + b × 40

Example calculation structure: If we had data points, we would:

1. Calculate means of X and Y
2. Calculate covariance and variance
3. Find slope (b) and intercept (a)
4. Form the prediction equation
5. Substitute X = 40 for prediction

Q55: Linear Regression - Glucose Level vs Age

Note: The specific table for glucose level and age is referenced but not provided in the PDF.

General Process:

Step 1: Data Setup

- Independent variable (X): Age
- Dependent variable (Y): Glucose level

Step 2: Regression Equation Following the same process as Q54: Glucose Level = a + b × Age

Step 3: Prediction for Age 55 Glucose Level = a + b × 55

Typical Process:

1. Plot the data points
2. Calculate correlation coefficient
3. Determine regression line
4. Validate the model
5. Make predictions

Q60: Multiple Linear Regression

Note: The specific dataset with response variable y and predictor variables X_1 and X_2 is referenced but not shown in the PDF.

Multiple Linear Regression Model: $Y = \beta_0 + \beta_1X_1 + \beta_2X_2 + \varepsilon$

Steps to construct the model:

Step 1: Data Preparation Organize data in matrix form with response variable Y and predictors X_1, X_2

Step 2: Calculate Regression Coefficients Using normal equations: $\beta = (X'X)^{-1}X'Y$

Where X is the design matrix including intercept term.

Step 3: Model Validation

- Check R-squared value
- Analyze residuals
- Test significance of coefficients

Step 4: Final Equation $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

Example interpretation:

- β_0: Intercept (value of Y when $X_1 = X_2 = 0$)
- β_1: Change in Y for unit change in X_1 (holding X_2 constant)
- β_2: Change in Y for unit change in X_2 (holding X_1 constant)

Q62: Fuzzy C-Means Clustering

Given Data Points: {(1, 3), (2, 5), (4, 8), (7, 9)} **Note:** Initial membership values are mentioned but not provided in the PDF.

Fuzzy C-Means Algorithm:

Step 1: Initialize Membership Matrix For 2 clusters and 4 data points, we need initial membership values u_ij where:

- i = cluster index (1, 2)
- j = data point index (1, 2, 3, 4)
- Σu_ij = 1 for each data point

Step 2: Calculate Cluster Centers For cluster k: v_k = Σ(u_ik^m × x_i) / Σ(u_ik^m)

Where m is the fuzziness parameter (typically m = 2)

Step 3: Update Membership Values u_ij = 1 / Σ[(d_ij/d_ik)^(2/(m-1))]

Where d_ij is the distance between data point j and cluster center i.

Step 4: Repeat Until Convergence Continue steps 2-3 until membership values stabilize.

General Process:

1. Initialize membership matrix
2. Calculate cluster centers
3. Update membership values
4. Check convergence criteria
5. Repeat until convergence

Q64: Logistic Regression - Pass/Fail Prediction

Given Information:

- Dataset: Pass/fail exam results for 5 students
- Model: Log(odds) = -64 + 2 × hours
- Questions: a) Probability of pass for 33 hours of study b) Hours needed for 95% pass probability

Solution:

Part (a): Probability for 33 hours

Step 1: Calculate Log(odds) Log(odds) = -64 + 2 × 33 = -64 + 66 = 2

Step 2: Calculate Odds Odds = e^(Log(odds)) = e^2 = 7.389

Step 3: Calculate Probability P(pass) = odds / (1 + odds) = 7.389 / (1 + 7.389) = 7.389 / 8.389 = 0.881

Answer: The probability of passing for a student who studied 33 hours is 88.1%

Part (b): Hours for 95% pass probability

Step 1: Set up the equation For P(pass) = 0.95: 0.95 = odds / (1 + odds)

Step 2: Solve for odds 0.95(1 + odds) = odds 0.95 + 0.95(odds) = odds 0.95 = odds - 0.95(odds) = 0.05(odds) odds = 0.95 / 0.05 = 19

Step 3: Calculate Log(odds) Log(odds) = ln(19) = 2.944

Step 4: Solve for hours 2.944 = -64 + 2 × hours 2 × hours = 2.944 + 64 = 66.944 hours = 33.472

Answer: A student should study at least 33.5 hours to have a 95% probability of passing.